

From Intangible Heritage to Virtual Play: Bridging Immersive Experience and Social Play in VR

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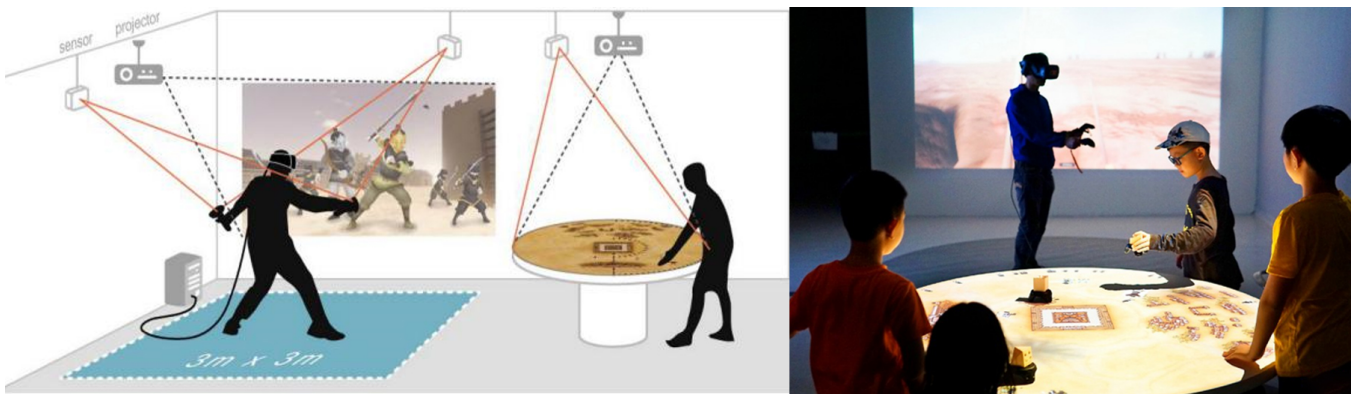


Figure 1: Schematic diagram of VR interactive game table and VR device linkage.

Abstract

This study presents Little Hero Wins the Masks, a multiplayer VR experience combining an interactive game table and a multiplayer VR environment. The game synchronized objects and events between the two platforms in real time, allowing players to collaboratively create immersive experiences. Through the user study, we analyzed the player interaction and experience in VR games with social elements. The goal is to offer insights for future VR development and explore ways to enhance the sociability of VR games. By integrating these interactive elements, the research aims to contribute to the development of more engaging and socially immersive VR games. The findings provide practical guidance for developers seeking to design VR systems that foster real-time, shared interactions among players while addressing technical challenges in cross-platform synchronization.

CCS Concepts

• **Human-centered computing** → *Empirical studies in interaction design.*

Keywords

GVirtual Reality, Sociability, Co-location Game, Interactive Table, Augmented Reality, Intangible Cultural Heritage.

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1 Introduction

For the past few years, virtual reality (VR) technology has gradually developed and matured, and more and more VR devices and software have been found on the market. Among VR related devices. The field with the strongest consumption capacity and largest scale is VR gaming. VR games and products are the design trends. Immersion is the biggest advantage of VR and users can immerse themselves in the virtual world. A large amount of studies have

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proposed that sociability is one of the most important factors in game experience[12, 19]. However, because of technical limitations, VR games are still mostly single-player experiences, and there are fewer designs that many people play together. Metz [11] believes that VR, which can only be experienced by only one user, is a big regret. Single VR will make users feel alienated. As long as the sociability is added, users can share the experience with others on VR and it will become the basis of a new communication tool. In the past decades, multi-user VR has been limited to laboratories or public places, because multi-user VR requires high-level display technology, computer graphics and network performance and environment,[10]. Now, the VR users are less. Due to the expensive hardware costs, VR cannot be widely used in our life, while HTC Vive and Oculus Rift, the two platforms with the most multi-user VR games, are almost completely independent of each other. This means that players cannot fight against each other across platforms (although HTC Vive supports Oculus Rift games to some extent, the support effect is not obvious). Therefore, at the end of 2017, we developed the game "Little Hero Wins the Masks" applying innovative virtual reality (VR) and interactive game table technology. It is a VR work adapted from an early Taiwanese comic, produced as part of a digitized Intangible Cultural Heritage project [7]. The game overcame existing technical bottlenecks, transforming VR games from being limited to single users to supporting multiplayer interactions through the development of various design tools. This study takes "Little Hero Wins the Masks" as an example to analyze the interactive taste and experience that VR games bring to users after adding sociability elements [7]. Hopefully it can give future VR developers a reference and put forward suggestions on the development direction of increasing the sociability of VR games. Our study shows that the interactive tables can significantly improve the sociability of VR players. After adding the interactive table, VR players have changed significantly in both the competence and challenge. While there was no significant change in the Sensory and Imaginative Immersion, flow, positive affect, negative affect and tension. Our study shows that the interactive table design is effective and will not degrade the game experience of VR players.

2 Related Works

In all ages, games are a part of human experience, and people also have created a variety of game activities for their own development. VR games and products have also become a design trend in recent years. Many scholars and game designers have pointed out that social interaction is very important for players to enjoy themselves, and sociability is an important part of the game experience[2–4]. Collaboration in social games improves social contact and happiness among players, creating supportive online environments that are gradually spilling over into real-world behavior. The Social Networking Service is infectious, and pleasant players are attracted to other pleasant players, which eventually leads to a spiral escalation of happiness in the social network. Sociability represents a conceptual criterion of system design in the field of human-computer interaction research [13], which mainly focuses on how people interact with others through computer systems [16], and attaches importance to how system design supports social relations between users. Compared with the traditional human-computer interaction

research topic, more importance is attached to the context and the way the system design supports the social relationship between users [7, 13]. Costikyan [5] believes that social interaction refers to a two-way communication mode with at least two or more people sharing the same purpose. To promote users' immersion in the game, the higher the degree of social interaction is the better. At present, VR games are mostly a single-player experience due to technical limitations, and there are fewer designs that many people play together. The small user base, expensive hardware costs, network technology and environment restrictions have all affected the development of multi-user VR games.

Traditional games are generally screen games, where players observe the game world from a view of "camera". The frame of the game is fixed, and the size of the vision field is limited by the screen size of multimedia devices (mobile phones, computers, tablet PC, etc.), so the immersion of the game is not strong. Virtual reality games don't have any screen restrictions, and players can see the whole world directly through their eyes by combining themselves and the roles in game into one through wearable devices. Therefore, the perspective of the game will be broader, and the immersion of the game will be stronger[8]. Many of the previous articles mentioned that the relationship between social games and immersion is controversial. When you are immersed in a game, the real people in the game will be regarded as the influence and interference of personal immersion. Of course, in some cooperative games, for example, if there is no interaction and cooperation between the players, then the game does not have much game experience. Certainly, in some games, interactions may be inherent and set, so the game experience is of great importance.

Since the original CAVE (Cave automatic virtual environment) system, co-location has always been a key feature of VR. In a VR system, only one user can interact, and other users are bystanders who can only watch through the monitor. Such co-location VR lacks social contact. Even if two or more CAVE systems are connected, co-location users can only passively obtain experience [10]. However, Maurice Benayoun [1] proposed a unique concept in the work of World Skin, A Photo Safari in the Land of War to achieve a co-location experience. In this work, up to three viewers can interact with real cameras in a virtual environment. It is also believed to be the first co-location work because of the combination of the virtual environment and the real world. In recent years, with the continuous development and progress of science and technology, mixed reality has also begun to develop. MR combines the real environment with the virtual world to create a new environment that conforms to the virtual image generally recognized in the general visual sense. Also, users coexist with each other and interact with the virtual image.

Co-position games create engaging social interactions, and these players' social interactions while playing the game contribute to the continued success of co-position games[9, 14]. In a VR environment, one aspect of the co-location is a phenomenon called presence. Generally speaking, in the virtual environment, existence mainly refers to the "sense of existence" of an individual in the virtual environment [17]. Witmer and Singer[20] believe that participation and immersive experience are requirements for existence, and a high degree of participation and immersion in a virtual environment

can improve the sense of existence. In addition to conceptualizing existence as "being there", researchers further conceptualize coexistence as a sense of coexistence. The concept of coexistence originates from Goffman[6] and is defined as a sense of living together in a virtual environment. Biocca[13] et al. believe that the definition of coexistence emphasizes that the sense of existence in the same virtual place or environment is related to some attributes of the place. On the other hand, the coexistence seen by others not only exists in the same place, but also has the mutual consciousness of individuals, and emphasizes the sensory characteristics of the virtual environment. Coexistence contains two dimensions: having the feeling of other individuals, and others are actively perceiving us and becoming part of the group [6, 18]. At present, the VR market is booming and co-location is a key feature of VR. However, research on the social aspects of VR games is rare.

3 VR GAME DESIGN: LITTLE HERO WINS THE MASKS

The game is divided into virtual reality (VR) and interactive table. VR and interactive tables are connected in real time, and objects and events of the two games are controlled synchronously through the program. It can provide asymmetric interactive experience between one VR player and up to six interactive table players.

After wearing the VR helmet, the VR player will enter the virtual world of Little Hero Wins the Masks. The game passes if VR players find and defeat the Grinning Mask, and the additional points are scored for killing Masked Soldier and retrieving six lost pages of ancient books. VR players have a controller in each hand, which is a sword for wielding and slashing. In addition, because of the limitation of the field, we set the other controller to move. The player selects the desired position by pressing the button on the controller, and releases it to the selected position.



Figure 2: VR character modeling

Players of the interactive table can help the VR player through the props on the interactive table. The VR player can also see the objects controlled by the interactive table player in the virtual environment, and can communicate directly through the voice.

Players can also control the elements in the VR real game through the figure cards on the interactive game table, so that players who don't know each other can interact happily without embarrassment to achieve a social game experience. There are four props on the interactive table, which are: A Borrowed Sword by Scarecrow, a Kite Assassin, a Mine Artillery Vehicle and a Hole. For example, if the figure card of gunpowder mine car on the interactive game table is controlled, it will project a bomb detonation on the interactive game table, presenting a brilliant 2D blasting effect. In virtual reality, a 3D blasting effect will appear. Players in VR games may also be affected if they do not dodge. In addition, it is also possible to deliberately place the hole figure card on the ground near the image of Si Lang, and VR players will fall into the ground with a very deep hole if there is any carelessness. The Borrowed Sword by Scarecrow can guide the VR player to quickly find the Grinning Mask. Using the Kite Assassin can throw the bomb down in the sky, and can quickly kill and repel the Soldier. Through the research through design, the game develops a social mixed-reality interactive device, and analyzes the interactive tastes and experience that VR games bring to users after adding social elements.



Figure 3: VR scene setting

After the completion of the creation, the entity will be exhibited, and 15 people who have just experienced the game will be invited to conduct questionnaire surveys and interviews. The preliminary interview results show that most of the respondents think that the VR in this creation is easy to operate and stimulating, and through the interactive table, the experienced people can interact with each other and increase the sociability of the game. Respondents showed a high willingness to re-experience and increased interest in further learning of Zhu Ge Si Lang comics. Therefore, the development of social mixed interactive devices and the performance of Taiwanese local comics are helpful for attracting people's interest, and it is worthy of further discussion and analysis. Therefore, this study seeks to address the following research questions:

- (1) How effective is the integration of interactive tables in enhancing social interaction among VR players?
- (2) What impact does the addition of social elements have on the overall game experience of VR players?



Figure 4: Interactive table operation in the exhibition of NYCU Art Gallery

4 USER STUDY

This experiment was conducted using the game experience questionnaire and post-mortem interviews. All the subjects signed the experimental study participation consent form.

In order to achieve the purpose of the research, we used the "Little Hero Wins the Masks" as an experimental tool. The experiment used the Game Experience Questionnaire (GEQ) and the post-interview method.

4.1 Experimental participant

This experiment recruited 30 subjects from students of National Yang Ming Chiao Tung University and National Tsing Hua University in Taiwan through the Facebook club, including 13 women (43.3%) and 17 men (56.7%), 18-23 years old and 18 people (60%), 24-30 years old (40%). 10 of them (33.3%) had VR experience, and the remaining 20 (66.6%) had no VR experience.

4.2 Experiment Facilities

In order to have a good visual effect and high performance experience of VR, the experimental facilities are HTC VIVE pro and a dedicated PC for VR, external links to a 23 inch ASUS display (synchronous display VR players screen). The screen is projected from the top to the bottom on the table with an ASUS projector. The four objects controlled by the interactive table player are HTC VIVE mobile locators.

4.3 Experimental Process

All subjects will sign an experimental consent form before the experiment. The researcher introduces the experimental background and process to the subject. The whole experiment is divided into two rounds. Each round has three players (one player is the subject and the other two are fixed staff). The first round of experiments: the subjects played VR games alone – Little Hero Wins the Masks, in three minutes, the other two players (staffs) watch the monitor next to it. The second round of experiments: add the interactive table, the subjects played VR games – Little Hero Wins the Masks, the other two players (staffs) play interactive table, through the

interactive table's four props to help players win the game.

The subjects conducted two rounds of experiments (with interactive table/no interactive table). The two experimental sequences were randomly assigned to ensure no adaptive learning and fatigue effect interference. 15 subjects first experienced an version with interactive table, and 15 subjects first experienced an version without interactive table. In order to avoid the interference of external factors, among the 30 groups of subjects, the two players of the interactive table are fixed two staff members. The difficulty, operation mode, time and winning conditions of the game were the same in both rounds.



Figure 5: Left: VR group . Right: interactive table group (with interactive table)

Subjects need to fill out the Game Experience Questionnaire (GEQ) after two rounds of experimentation. The Game Experience Questionnaire (GEQ) is a questionnaire widely used in various game research in recent years[15]. The questionnaire was developed by the European research project FUGA and was divided into three modules: GEQ, the post-game experience scale and Social Presence Module. In this experiment, we used the In-Game GEQ (IGE) and the Social Presence Module (SPM). The In-Game GEQ, the short version of the GEQ, is an extra-simplified scale for GEQ compilers to evaluate continuous, repetitive evaluations over a long game experience. The In-Game GEQ consists of seven dimensions: Sensory and Imaginative Immersion, flow, ability, positive affects, challenges, negative affects, and tension. Each dimension has two questions. The Social Presence Module is divided into three dimensions: Psychological Involvement - Empathy, Psychological Involvement - Negative Feelings, and Behavioural Involvement. Psychological Involvement - Empathy, consists of six questions that represent the player's perception and understanding of other players. There are 5 questions about negative affects. Negative affects used to refer to negative and bad emotions among players. However, in this experiment, because of the design of the hole of interactive table props (interactive table players can be used to frame VR players), it is a negative affect with a playful nature in this experiment. There are 6 questions in Behavioural Involvement. It mainly refers to the degree to which players participate in the game of personal involvement. The In-game GEQ and the Social Presence Module are all five-point scales. Players evaluate the degree of recognition for each topic from 0 (totally disagree) to 4 (completely agree). Because each subject has two rounds of experiments, the corresponding game experience questionnaire GEQ (with interactive table / without interactive table) and social presence scale (with interactive table / without interactive table) will be filled out separately. Finally, after

two rounds of experiments, simple one-on-one interviews will be conducted to further learn the details and feelings of the subjects in the experiment.

5 Result

For 30 students between the ages of 19 and 30, the game experience and social test of Little Hero Wins the Masks with or without interactive tables are conducted. Players' Behavioural Involvement, Psychological Involvement - Empathy, and Psychological Involvement - Negative Feelings are all significantly different from those without interactive table. In the Behavioural Involvement, the interactive table model ($M=2.83$, $SD=0.49$) was significantly higher than the model without interactive table ($M=1.13$, $SD=0.99$), $t(30)=8.03$, $p<0.05$, $d=1.47$. It shows that after the addition of interactive table, the interactive table player and the VR game player interact with each other. In the Psychological Involvement - Empathy, the mode of adding interactive table ($M=2.04$, $SD=0.53$) is also significantly higher than that without interactive table ($M=1.00$, $SD=0.92$), $t(30)=6.13$, $p<0.05$, and $d=1.12$. Under the mode of interactive table, VR players and interactive table players are closely connected, and VR players and interactive table players feel happy to cooperate. In the orientation of negative emotions, the pattern of interactive table ($M=1.30$, $SD=0.67$) is still higher than that without interactive table ($M=0.78$, $SD=0.66$), $t(30)=6.20$, $p<0.05$, $d=1.13$. It may be because the hole in the interactive table is two-way, which can help the VR player to kill the Soldier or frame the VR player and drop it into the hole, so the player of the interactive table can influence the emotions of VR players. VR players want to "revenge" interactive table players instead of real negatives orientation. This conclusion needs further discussion. Therefore, the three aspects of the comprehensive social presence scale can be found that after adding the interactive table, the sociability of the VR game has been significantly improved, thus verifying our initial guess - the addition of the interactive table is effective to improve the sociability of VR player.

In terms of game experience, the presence or absence of interactive table is significant in competence and challenge. Subjects believed that the competence was significantly higher than that of the non-interactive table ($M=2.21$, $SD=0.72$), $t(30)=3.03$, $p<0.05$, in the case of interactive tables ($M=2.58$, $SD=0.68$), $d=0.55$. After the addition of the interactive table, because the cooperation makes it easier for the player to win the game, the player will feel more fulfilling and skillful in the interactive table mode. On the challenging side, VR games with interactive tables ($M=2.23$, $SD=0.87$) were significantly higher than those without interactive tables ($M=1.98$, $SD=0.73$), $t(30)=2.4$, $p<0.05$. In the case of an interactive table, because of the need to cooperate and socialize, the player will think that the game is more challenging and thus put more effort into the game. In terms of positive affects, negative affects, uneasiness, irritability, Sensory and Imaginative Immersion and flow, there is no significant difference in the presence or absence of interactive tables. This also answers our initial question 2. After addition of the sociability, the competence and challenge have changed significantly.

In the past research, most of the researches believe that Sensory and Imaginative Immersion is an important part of the game experience, so as to discuss the relationship between sociability and Sensory

and Imaginative Immersion, without mentioning other aspects such as competence and challenge. This study finds that because of the addition of interactive tables, it has become social, and because of the increase in sociability, the players have obvious differences improvement in the two dimensions of competence and challenge. In the follow-up interviews, text analysis revealed several key concepts frequently mentioned by players. Many described the game as an *experience of Sensory and Imaginative Immersion* (U2, U5, U8-11, U18), while others highlighted that *the cooperative aspect of the interactive table was particularly engaging* (U1, U4, U11-14, U16, U22-25). Players also found it entertaining to *use the hole to "trap" others*, as it not only helped VR players eliminate enemies but also allowed them to cause other players to fall into the hole (U4, U13-14, U23). Furthermore, players emphasized that *the presence of other players was strongly felt in the interactive table mode*, enhancing the sense of sociability (U1-8, U10-16, U18, U20-25, U27-29). Additionally, elements like *scarecrows and holes were noted as being more noticeable*; scarecrows and holes were noted as being more noticeable in this mode (U3, U5, U10-12, U22).

Overall, most participants appreciated the interactive table's ability to enrich VR gameplay and enhance social interaction.

6 CONCLUSION AND FUTURE RESEARCH PLANNING

Our experiment demonstrates that the integration of interactive tables effectively enhances the sense of social presence among VR players, addressing a key challenge in VR game design by transforming the one-person of VR gaming into a more collaborative experience. This corresponds to the first research question, showing that interactive tables can significantly improve sociability in VR gaming environments. Regarding the second research question, the inclusion of social elements, such as the interactive table, notably enhances player experiences in terms of competence and challenge without negatively impacting positive affect, negative affect, irritability, sensory immersion, imaginative immersion, or flow. Players also expressed increased interest in learning about the cultural and narrative aspects of Little Hero Wins the Masks after gameplay, underscoring the potential of sociable VR games to foster cultural engagement. Future work will focus on optimizing the interactive table's design to enhance gameplay fluidity and identifying additional factors that influence sociability in VR games. These findings aim to provide actionable design recommendations for VR developers and explore new directions for incorporating social elements into VR gaming.

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